

Kenneth Watts

Philadelphia, PA, 19104
+1 (445) 310 0675

kenwatts@wharton.upenn.edu

Education

University of Virginia, Incoming PhD in Statistics August 2026 – May 2030

Research interests: Applied Probability, Dependence Modeling, Sports Analytics and Bayesian methods

University of Pennsylvania, Bachelor of Science in Economics August 2022 – May 2026

Concentrations: Statistics and Quantitative Finance

GPA: 3.75/4.00

GRE: 330/340 (170 Quantitative Reasoning and 160 Verbal Reasoning)

Takapuna Grammar School, New Zealand January 2020 – November 2021

Dux (Valedictorian Equivalent)

SAT: 1590/1600

Work Experience

Moneyball Academy, University of Pennsylvania, Wharton June 2025 – August 2025

Teaching Assistant

Teaching assistant for Professor Wyner's Moneyball Academy, teaching students about statistics through sports examples using the R programming language

University of Pennsylvania, Wharton August 2023 – May 2025

Teaching Assistant

Teaching assistant for Professor Gupta's STAT 1020 Course, grading homeworks and exams, and proctored midterms

University of Pennsylvania, Weingarten Center August 2023 – May 2024

Peer Tutor

Provided one-on-one and group tutoring for undergraduate corporate finance courses.

First Cape, Global Equities, Auckland May 2024 – August 2024

Equity Research Analyst

Sole analyst under the Global Equities Portfolio Manager for a \$1.5bn portfolio.

Led meetings with sell-side analysts and investor relations.

Built valuations models for 25 portfolio companies and updated a quantitative screening tool for stock selection.

Jarden, Wealth Research, Auckland June 2023 – August 2023

Equity Research Analyst

Built an interactive client-portfolio dashboard using Power BI

Cleaned and analyzed large Excel data sets to assist with portfolio diversification and optimization

Research Experience

Injury modeling for the Cleveland Browns December 2025 – Present

Streamlining data feeds relating to player injuries to calculate, and then predict, games missed due to injuries.

Creating NFL player comparisons, in terms of injury profile, for pre-draft college football players. Selected as one of 6 undergraduate/graduate students from the Wharton Sports Analytics and Business Initiative to complete this project.

Opponent-Adjusted Evaluation of NFL Pass Blocking and Pass Rushing Performance

Developed a framework for analyzing the quality of NFL offensive and defensive linemen based on player tracking data and individual play interactions

Used a ridge-regularized Bradley-Terry model aligned with a four-class severity model over loss/win/hit/sack outcomes

Selected for a poster presentation at the New England Symposium on Statistics in Sports (NESSIS) at Harvard University

Win Probability in Rugby Union

December 2025 – Present

Making the first win probability paper in Rugby Union, using a First-order Markov chain approach. Paper to be submitted soon.

Evaluating the Effect of Substitutions on Soccer Scoring Dynamics

November 2025 – Present

Assisting Professor Paul Sabin and PhD student Kaishu Mason with making changes for resubmission on their paper on modeling optimal substitution timings in soccer

Kicking for Goal or Touch? An Expected Points Analysis in Rugby Union

September 2025

– November 2025

Designed and implemented the first ever application of an Expected Points framework for the sport of Rugby Union

Developed an expected points surface evaluating the decision between a kick at goal and going for a lineout (Similar to fourth down decision in NFL), depending on field location allowing for the comparison between decisions at a specific penalty's coordinates

Submitted to JQAS (Journal of Quantitative Analysis in Sports)

Arxiv Link for the paper

Poster Presentations

A Dynamic Rating System for NFL Pass Rushers and Blockers

New England Symposium on Statistics in Sports (NESSIS), Boston

September 2025

Wharton Sports Business Summit, Philadelphia

September 2025

Awards & Achievements

United States National Collegiate Rugby All Star

2024/25

University of Pennsylvania Rugby Captain

2023/24 and 2024/25

New Zealand Top Scholars Award – Takapuna Grammar School

2021

Dux (Valedictorian Equivalent) – Takapuna Grammar School

2021

Miscellaneous

Reading: Have read 35–40 books per year since 2021

Some favorites include: Fooled By Randomness, The Billion Dollar Molecule, Barbarians at the Gate, Catch-22, When Genius Failed, and the Jason Bourne Books.

Running: Sub 17 minute 5k; run 3–4x per week.
